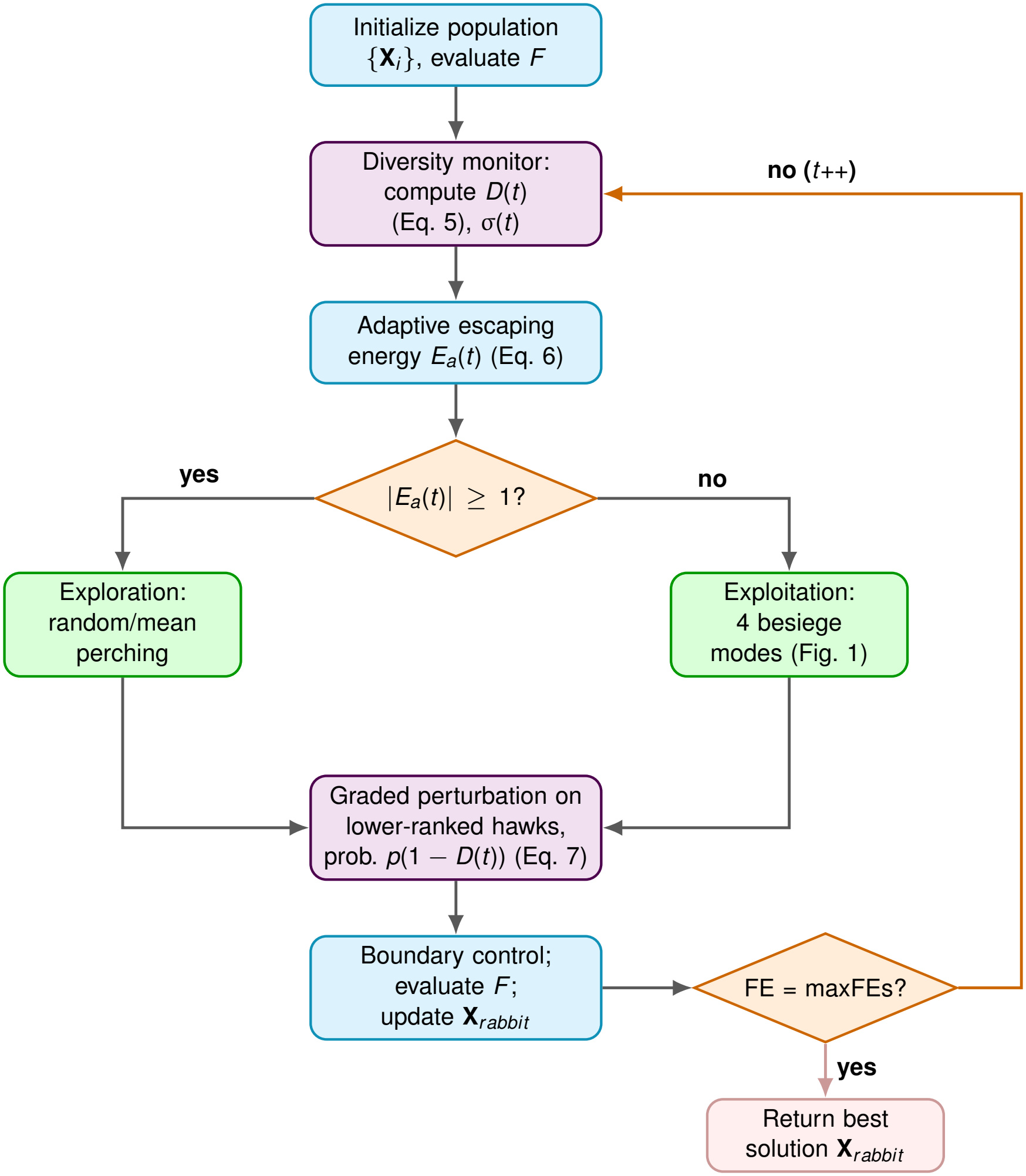
Initialize population $\{\mathbf{X}_i\}$, evaluate F
Diversity monitor: compute $D(t)$ (Eq. 5), $\sigma(t)$
no (t++)
Adaptive escaping energy $E_a(t)$ (Eq. 6)
$|E_a(t)| \geq 1$?
yes
no
Exploration: random/mean perching
Exploitation: 4 besiege modes (Fig. 1)
Graded perturbation on lower-ranked hawks, prob. $p(1 - D(t))$ (Eq. 7)
Boundary control; evaluate F; update $\mathbf{X}_{rabbit}$
FE = maxFEs?
yes
Return best solution $\mathbf{X}_{rabbit}$